AP PRECALCULUS

Unit 1 • Worksheet 1.12MEGA • Answer key

Worksheet 1.12MEGA • Transformations of Functions

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

$g(x) = f(x-2) + 5$. Describe the transformation.

(A) left 2, up 5
(B) right 2, up 5
(C) right 5, up 2
(D) left 5, down 2

Solution

Answer: (B).

Question 2

NO CALCULATOR

$g(x) = \frac{1}{3}f(x)$. Describe the transformation.

(A) vertical dilation by $\frac{1}{3}$
(B) vertical dilation by 3
(C) horizontal dilation by $\frac{1}{3}$
(D) horizontal dilation by 3

Solution

Answer: (A).

Question 3

NO CALCULATOR

$g(x) = f\left(\frac{1}{2}x\right)$. Describe the transformation.

(A) horizontal dilation by $\frac{1}{2}$
(B) vertical dilation by 2
(C) horizontal dilation by 2
(D) vertical dilation by $\frac{1}{2}$

Solution

Answer: (C). Factor $\frac{1}{1/2} = 2$.

Question 4

NO CALCULATOR

Which function is the graph of $y = |x|$ reflected over the x-axis and shifted right 3?

(A) $-|x+3|$
(B) $|-x-3|$
(C) $-|x|+3$
(D) $-|x-3|$

Solution
Answer: (D).

Question 5

NO CALCULATOR

The vertex of $y = x^2$ moves to $(-1, 4)$ and the graph opens downward with the same shape. Which is the equation?

(A) $y = -(x+1)^2 + 4$
(B) $y = -(x-1)^2 + 4$
(C) $y = (x+1)^2 - 4$
(D) $y = -(x+4)^2 - 1$

Solution
Answer: (A).

Question 6

NO CALCULATOR

$(6, 2)$ is on the graph of f. Which point is on $y = f(3x)$?

(A) $(18, 2)$
(B) $(6, 6)$
(C) $(2, 2)$
(D) $(2, 6)$

Solution
Answer: (C). $x = \frac{6}{3} = 2$.

Question 7

NO CALCULATOR

f has range $[1, 5]$. What is the range of $y = -f(x) + 2$?

(A) $[-3, 1]$
(B) $[3, 7]$
(C) $[-5, -1]$
(D) $[1, 5]$

Solution
Answer: (A). $-f \in [-5, -1]$, then $+2$.

Question 8

NO CALCULATOR

Which transformation can change the domain of a function?

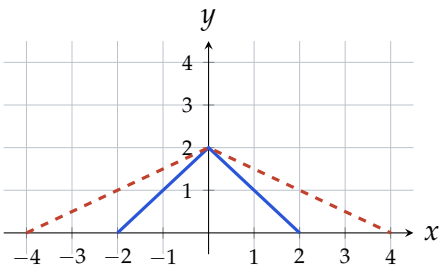
(A) $f(x)+4$
(B) $3f(x)$
(C) $-f(x)$
(D) $f(x-4)$

Solution
Answer: (D). Horizontal changes affect the domain.

Question 9

NO CALCULATOR

The graph of f is shown (solid). The dashed graph is g. Which is g?



(A) $f(2x)$
(B) $2f(x)$
(C) $f\left(\frac{1}{2}x\right)$
(D) $\frac{1}{2}f(x)$

Solution
Answer: (C). Twice as wide, same height: horizontal dilation by 2.

Question 10

NO CALCULATOR

Which describes $g(x)=f(-x)$ for an even function f?

(A) $g=-f$
(B) $g=f$
(C) g is f shifted left
(D) g is undefined

Solution
Answer: (B). Even: $f(-x)=f(x)$.

Question 11 (Free response) NO CALCULATOR

Let $f(x) = x^2$ and $g(x) = -2(x-1)^2 + 8$.

(a) Describe the transformations from f to g.
(b) Find the zeros of g and its maximum value.
(c) If f is restricted to $[-2, 2]$, find the corresponding domain and range of g.

Solution
(a) Right 1, vertical dilation by 2, reflection over the x-axis, up 8.
(b) $(x-1)^2 = 4$: $x = 3$, $x = -1$. Maximum 8 at $x = 1$.
(c) Domain $[-1, 3]$; range $[0, 8]$.

Question 12 (Free response) NO CALCULATOR

The graph of f is the line segments through $(-2, 1)$, $(0, 3)$, $(2, -1)$. List the corresponding points on each graph: (a) $y = f(x) + 2$ (b) $y = f(x + 2)$ (c) $y = -f(x)$ (d) $y = 2f\left(\frac{x}{2}\right)$

Solution
(a) $(-2, 3), (0, 5), (2, 1)$. (b) $(-4, 1), (-2, 3), (0, -1)$. (c) $(-2, -1), (0, -3), (2, 1)$. (d) $(-4, 2), (0, 6), (4, -2)$.